

Case Study III: OptiKuh Dairy Cow Health Data

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What we did

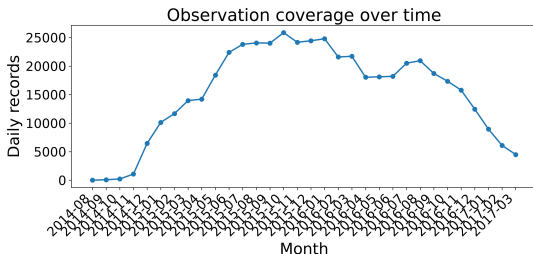
- ▶ Set up a reproducible project structure in VS Code
- ▶ Converted and organized the OptiKuh dataset for analysis
- ▶ Performed exploratory data analysis on:
 - ▶ observation coverage
 - ▶ health-status distribution
 - ▶ farm and breed differences
 - ▶ disease categories
 - ▶ missingness patterns
 - ▶ milk yield and metabolic markers
- ▶ Generated reusable figures, tables and notebook outputs

Overview and Preprocessing

- ▶ Loaded the original Excel data and metadata files
- ▶ Converted the main dataset into CSV format for faster repeated loading
- ▶ Parsed date variables and checked the observation period

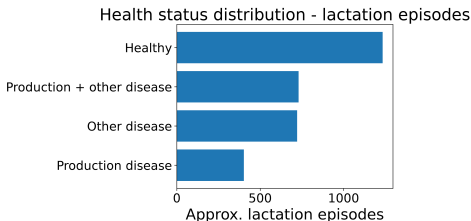
Analysis 1.1

- Observation coverage changes strongly over time
- The dataset is longitudinal, not one independent row per cow
- This affects how future modelling data should be split



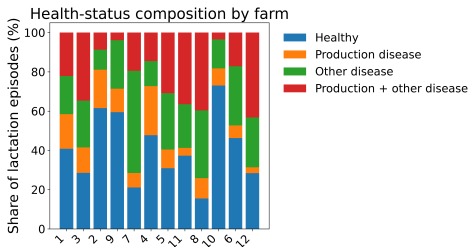
Analysis 1.2

- ▶ Health status was summarized at lactation-episode level
- ▶ This avoids repeatedly counting the same animal across daily records
- ▶ Healthy episodes form the largest group, but disease categories are still substantial



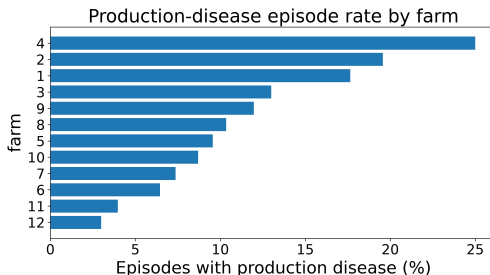
Analysis 1.3

- Health-status composition differs clearly across farms
- Some farms have a much larger share of disease-related episodes
- Farm identity should therefore be considered in later validation



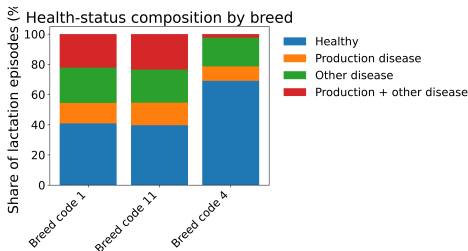
Analysis 1.4

- Production-disease rate is not evenly distributed across farms
- This is important before deciding on train-test split strategy
- A random row-wise split could give overly optimistic model performance



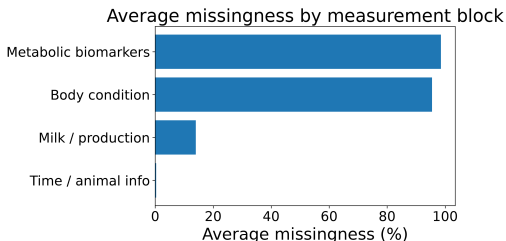
Analysis 1.5

- ▶ Breed groups show different health-status distributions
- ▶ Breed may influence both disease risk and production-related variables
- ▶ This should be retained as a grouping variable in future analysis



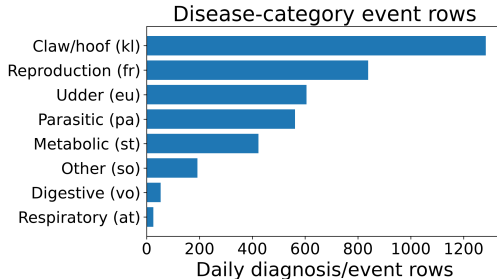
Analysis 1.6

- ▶ Missingness is strongly measurement-dependent
- ▶ Biomarkers and body-condition variables are much sparser than basic animal/time variables
- ▶ Missing biomarker values should not be treated as simple random missingness



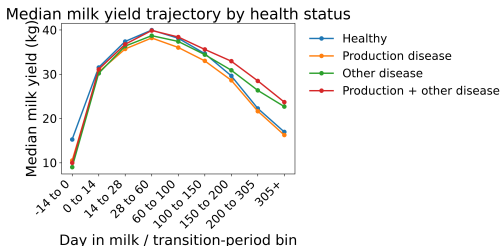
Analysis 1.7

- Disease events are concentrated in a few main categories
- Claw/hoof, reproduction and udder-related disorders dominate the recorded events
- These categories are important for defining future target variables



Analysis 1.8

- Milk yield changes over days in milk and differs by health status
- The time axis is central for interpreting production-related patterns
- Future features should therefore consider lactation-stage windows



Agenda for next week

- ▶ Define the final analysis unit:
 - ▶ daily record vs. lactation episode
- ▶ Define the production-disease target clearly
- ▶ Decide the time window around calving
- ▶ Build preprocessing steps for sparse biomarker variables
- ▶ Start baseline modelling only after the target and split strategy are fixed